2024 Calculus II Additional Exercise (7)

1. Exercise 4 in page 38 from Calculus Supplement Homework.

2. (1). Find the unit tangent vector $\mathbf{T}$ and principal mormal vector $\mathbf{N}$ for the plane curve C: $y = x^3$.
(2). Find the curvature function $\kappa(x)$ for the curve C and find points at which $\kappa(x)$ has respectively its absolute maximum and minimum values.
(3). Find the osculating circle for the curve C at $x = 1$.

3. Exercise 24, (1),(3),(5),(7),(9) in page 41 from Calculus Supplement Homework.

4. Exercise 42 in page 43 from Calculus Supplement Homework.

5. Let $f(x, y) = \frac{\sin xy}{\sqrt{x^2+y^2}}$. Define the value of $f(0, 0)$ such that f is continuous at $(0, 0)$.